WEEK 3 · 16–19 NOVEMBER 2026

Project workshop and interim quality assurance

Building the first defensible version of your project, then tearing it apart on purpose

Week 3 · Sessions 4–5 · practical: templates/project_template.ipynb

LEARNING OBJECTIVES

What you will be able to do

- Structure your project notebook so a reviewer can follow it without you in the room
- Run your own work through the interim QA checklist before anyone else does
- Give and receive peer review that improves a project instead of bruising it
- Recognise the five failure patterns that sink Earth observation projects
- Arrive at the interim review with the right four things ready

1. The workshop day

From scattered notebooks to one runnable project

THE WEEK 3 STANDARD

By the review you need a result you can defend, not a result you like.

A rough map with an honest accuracy figure beats a beautiful map with no validation. The rubric rewards the first and punishes the second.

A SUGGESTED TIMETABLE

How the workshop day runs

Time	What happens	Output
09:00–09:30	Stand-up: each team states its one blocking problem	Blockers listed
09:30–12:00	Build: config, data, method — instructors circulate	Notebook runs top to bottom
13:00–15:00	Validate: spatial split, metrics, baseline	First honest numbers
15:00–16:00	Self-QA with the checklist	Checklist filled in
16:00–17:00	Swap notebooks with another team, peer review	Review returned

Use the project template, in its order

- **1 Problem, user and decision** — copied from your scoping canvas, not reinvented
- **2 Configuration** — scene IDs, window, thresholds, seeds, all in one cell
- **3 Data and quality control** — quicklook, masks inspected, usable fraction stated
- **4 Preprocessing** — bad bands, and any extra step justified in one line
- **5 Features → 6 Method** — always with a baseline next to the main method
- **7 Validation → 8 Uncertainty and limitations** — the part reviewers read first
- **9 Product output → 10 Conclusions** — COG plus metadata, then what it means

What 'finished' looks like

- One command reproduces the map
- The output is georeferenced and has metadata
- A quicklook a non-specialist can read
- A number with a unit, and its uncertainty
- Quality flags that say when not to trust it





Example product output from notebook 31

2. Quality assurance

Check your own work before someone else does

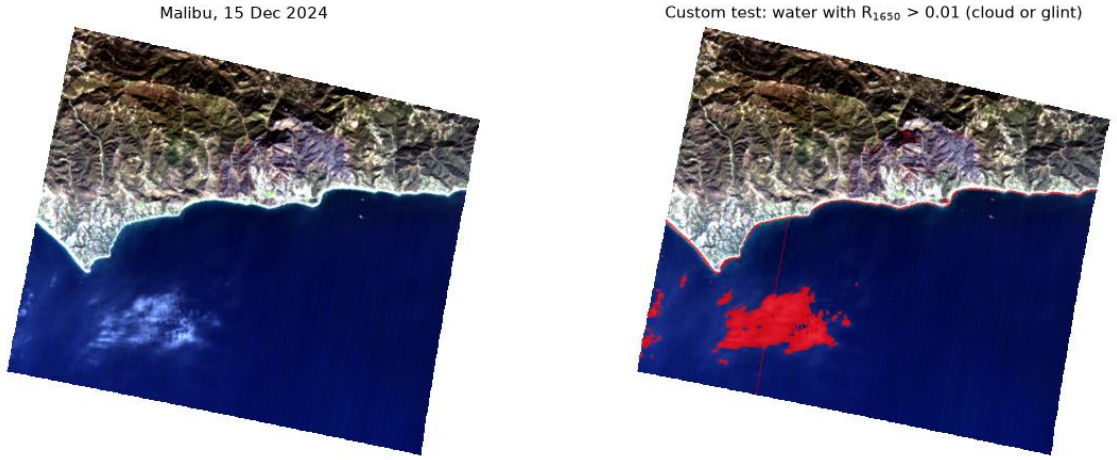
A — Reproducibility

- *Restart & Run All* finishes on a clean kernel — the single most common failure
- No absolute paths; data found through `hu.get_scene()` or a relative folder
- One config cell holds scene IDs, windows, thresholds and seeds
- Random seeds fixed: `np.random.default_rng(42)`, `random_state=42`, `torch.manual_seed`
- Runtime is sane, or heavy steps are cached and clearly marked

SECTION B

B — Data and preprocessing

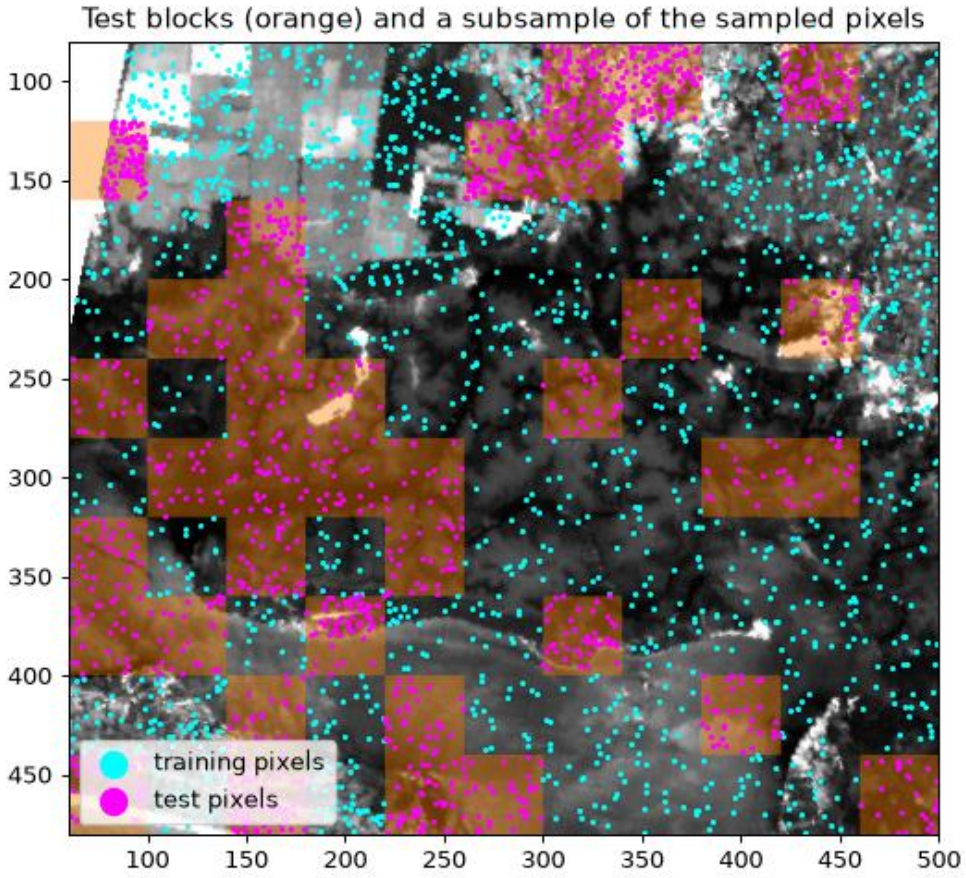
- Every dataset credited, including the reference labels
- QA masks applied **and inspected** — the sabkha flagged as cloud in Al Dhafra
- Bad bands removed, and the choice explained in one sentence
- Band choices state the wavelength used, e.g. 'NIR = 862 nm'
- Negative reflectance over water understood, not silently deleted



Inspecting the cloud mask before trusting it

C — Method and validity

- The method still matches the scoping canvas, or the change is explained
- The hyperspectral advantage is demonstrated or honestly discussed
- Train and test data are **spatially separated**
- Metrics come with the error matrix, not as a lone number
- A baseline shows what the complexity bought
- Failure cases shown, uncertainty communicated



A spatially separated train/test split (notebook 15)

D — Communication

- Every figure has a title, axis labels with units, colour bar and legend
- Colour maps chosen sensibly — no rainbow for continuous data, categorical colours for classes
- Markdown explains *why* before each major step, not just what the code does
- Key numbers gathered in one results table
- The user-facing product is sketched, even if only as a mock-up

3. Peer review

The cheapest way to find your own blind spots

How to give a review that helps

- Run the notebook **first**, before reading it — most findings come from that alone
- Mark each checklist item ✅ / ⚠ / ❌ and add one line of evidence for every ⚠ and ❌
- Separate **defects** (it does not run, the split leaks) from **preferences** (I would colour it differently)
- Lead with the one change that would most improve the project
- Be specific and kind: 'the test blocks touch the training blocks at row 200' beats 'validation is weak'

How to receive one

- Do not defend in the room — write the finding down and check it afterwards
- Anything that made the reviewer stumble is a real problem, even if you can explain it
- Triage into: fix now, fix before submission, document as a limitation
- Attach the completed checklist to your interim submission, including the ❌ items

RECOGNISE THESE EARLY

Five ways Earth observation projects fail

Pattern	Symptom	Fix
Leakage	Accuracy above 0.95 on a random pixel split	Blocked split, buffer, re-measure
No baseline	Only the complex model is reported	Add a threshold or index comparison
Unvalidated map	Beautiful figure, no numbers	Sample the map, report UA/PA with CIs
Map of nothing	Pretty output, no user or decision	Return to the scoping canvas
Cherry-picked scene	Works on one window only	Test on a second area, report both

4. The interim review

What to bring, and what happens

10 MINUTES PER TEAM, PLUS 5 FOR FEEDBACK

Each team brings four things

- The project notebook, executed, running top to bottom
- One map or result figure, and the numbers behind it
- The completed QA checklist, ❌ items included
- One question you want the panel to answer

What 'done' means by the end of week 3

- The notebook runs from a clean kernel on another machine
- There is a **first honest accuracy figure**, from a spatially independent design
- A baseline exists and the comparison is written down
- At least three failure cases are identified and explained
- The remaining work for week 4 is a list, not a hope

PRACTICAL

Over to you

▶ `templates/project_template.ipynb` · 180 min

- Restructure your work into the template's section order
- Fill in templates/interim_qa_checklist.md for your own notebook, honestly
- Swap with another team and review theirs against the same checklist
- Turn the findings into a prioritised fix list for week 4

KEY TAKEAWAYS

Recap

- The template's order is the reviewer's reading order — use it
- Self-QA before peer review; peer review before the panel
- Reproducibility failures are the cheapest marks to lose and the easiest to fix
- Name your failure pattern early: leakage, no baseline, unvalidated, no user, one scene
- Bring a defensible number to the review, not a beautiful picture

Next: Week 4 — Advanced analytics and solution refinement, then final implementation

References and further reading

templates/project_template.ipynb · templates/interim_qa_checklist.md (this course)

assignments/assignment_w3_validated_map.md — the interim milestone deliverable

Wilson, G. et al. (2017). Good enough practices in scientific computing. PLoS Comput Biol 13(6), e1005510.

Sandve, G.K. et al. (2013). Ten simple rules for reproducible computational research. PLoS Comput Biol 9(10), e1003285.

Rule, A. et al. (2019). Ten simple rules for writing and sharing computational analyses in Jupyter Notebooks. PLoS Comput Biol 15(7), e1007007.